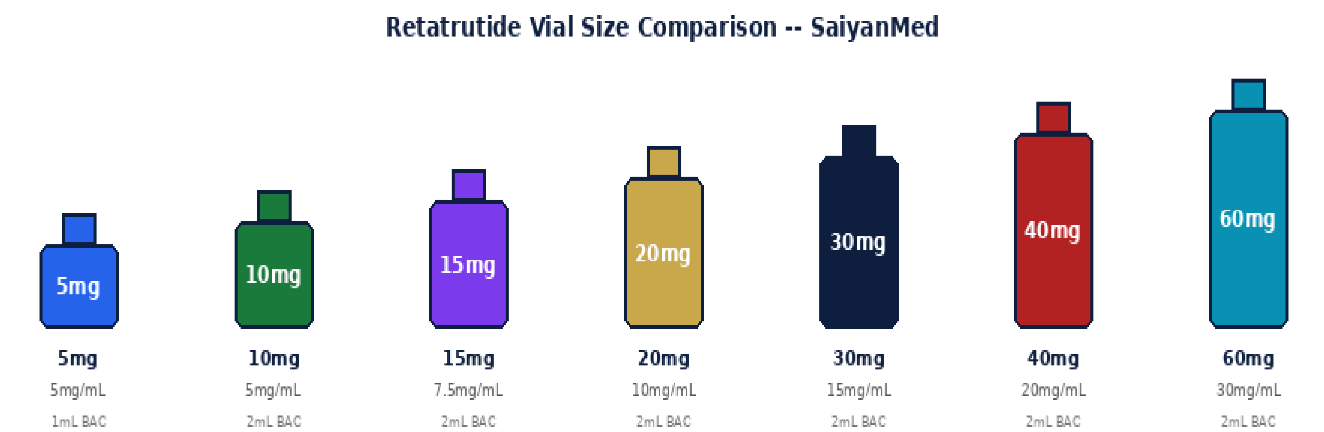
RETATRUTIDE

Triple GIP + GLP-1 + Glucagon Receptor Agonist

Complete Injection & Dosing Guide -- All Vial Sizes

Purity: 99.8% | Endotoxin-Free | Heavy Metal-Free | Research Use Only

Vial Size Comparison



Vial	BAC Water	Concentration	Doses/Vial	Dose Range per Kit (x10)
5mg x 10	1 mL	5.0 mg/mL	10	10 doses (0.5mg ea) | 5 doses (1mg ea)
10mg x 10	2 mL	5.0 mg/mL	20	20 doses (0.5mg ea) | 10 doses (1mg ea)
15mg x 10	2 mL	7.5 mg/mL	30	30 doses (0.5mg ea) | 15 doses (1mg ea)
20mg x 10	2 mL	10.0 mg/mL	40	40 doses (0.5mg ea) | 20 doses (1mg ea)
30mg x 10	2 mL	15.0 mg/mL	60	60 doses (0.5mg ea) | 30 doses (1mg ea)
40mg x 10	2 mL	20.0 mg/mL	80	80 doses (0.5mg ea) | 40 doses (1mg ea)
60mg x 10	2 mL	30.0 mg/mL	120	120 doses (0.5mg ea) | 60 doses (1mg ea)

WHAT IS RETATRUTIDE?

Retatrutide is a **next-generation triple receptor agonist** that simultaneously activates three metabolic pathways: **GIP** (Glucose-dependent Insulinotropic Polypeptide), **GLP-1** (Glucagon-Like Peptide-1), and **Glucagon** receptors. This tri-agonism produces superior weight reduction, metabolic improvement, and enhanced energy expenditure compared to single or dual agonists.

Receptor	Primary Action	Research Benefit
GIP	Insulin sensitisation, fat storage regulation, bone metabolism	Amplifies weight loss; reduces GI side effects vs GLP-1 alone
GLP-1	Appetite suppression, gastric emptying delay, insulin secretion	Reduces caloric intake; improves glycaemic control
Glucagon	Hepatic glucose output, thermogenesis, lipolysis	Enhanced fat mobilisation and energy expenditure

Key Research Benefits

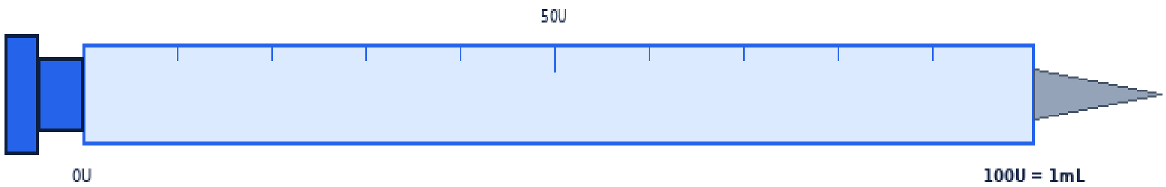
Benefit	Research Context
Superior Weight Reduction	Phase 2 research indicates potential for 24%+ body weight reduction over 48 weeks, exceeding results observed with GLP-1 or dual-agonist compounds.
Tri-Agonist Synergy	Simultaneous activation of three receptor pathways produces additive effects on appetite regulation, fat metabolism, and thermogenesis.
Improved Glycaemic Control	GLP-1 and GIP receptor activation improves insulin secretion and sensitivity, relevant in insulin-resistant research models.
Hepatic Fat Reduction	Glucagon receptor agonism promotes hepatic fat clearance; potential relevance to NAFLD and NASH research.
Lean Mass Consideration	Tri-agonism may help preserve lean muscle mass relative to caloric restriction alone; an active area of ongoing research.
Cardiovascular Markers	Research models show improvements in blood pressure, lipid profiles, and inflammatory markers alongside weight reduction.

RECONSTITUTION PROTOCOL

Always use **Bacteriostatic Water (BAC Water)**. Allow the peptide vial and BAC Water to reach room temperature before mixing. Use a fresh insulin syringe for each reconstitution.

Step	Action	Detail
1	Gather supplies	Peptide vial, BAC Water vial, insulin syringes (29-31G), alcohol swabs, sharps container, date labels.
2	Clean vial stoppers	Wipe stoppers of BOTH vials with a fresh alcohol swab. Air-dry for 10-15 seconds.
3	Draw BAC Water	Draw the correct volume per vial size (Page 1 table). Expel air bubbles before proceeding.
4	Inject into peptide vial	Direct stream against the GLASS WALL — not onto the powder. Inject slowly and steadily.
5	Mix gently	Roll vial between palms 20-30 seconds. NEVER shake — agitation degrades the peptide.
6	Inspect solution	Must be clear and colourless. Discard if cloudy, discoloured, or contains particles.
7	Label and refrigerate	Write reconstitution date on vial. Store at 2-8 deg C. Use within 28 days. Never freeze.

DOSE CALCULATION GUIDE



Insulin Syringe -- 1mL / 100 Units (29-31G needle)

Formula: Volume (mL) = Desired Dose (mg) / Concentration (mg/mL) | **Syringe Units:** Volume (mL) x 100 = Units on insulin syringe

Dose Escalation Ladder -- Retatrutide

Vial: 5mg x 10 — Add 1 mL BAC Water → Concentration: 5.0 mg/mL

Dose (mg)	0.5mg	1.0mg	2.0mg	4.0mg	6.0mg	8.0mg
Volume (mL)	0.100	0.200	0.400	0.800	1.200	1.600
Syringe Units	10.0U	20.0U	40.0U	80.0U	120.0U	160.0U

Vial: 10mg x 10 — Add 2 mL BAC Water → Concentration: 5.0 mg/mL

Dose (mg)	0.5mg	1.0mg	2.0mg	4.0mg	6.0mg	8.0mg
Volume (mL)	0.100	0.200	0.400	0.800	1.200	1.600
Syringe Units	10.0U	20.0U	40.0U	80.0U	120.0U	160.0U

Vial: 15mg x 10 — Add 2 mL BAC Water → Concentration: 7.5 mg/mL

Dose (mg)	0.5mg	1.0mg	2.0mg	4.0mg	6.0mg	8.0mg
Volume (mL)	0.067	0.133	0.267	0.533	0.800	1.067
Syringe Units	6.7U	13.3U	26.7U	53.3U	80.0U	106.7U

DOSE CALCULATION GUIDE (CONTINUED)

Vial: 20mg x 10 — Add 2 mL BAC Water → Concentration: 10.0 mg/mL

Dose (mg)	0.5mg	1.0mg	2.0mg	4.0mg	6.0mg	8.0mg
Volume (mL)	0.050	0.100	0.200	0.400	0.600	0.800
Syringe Units	5.0U	10.0U	20.0U	40.0U	60.0U	80.0U

Vial: 30mg x 10 — Add 2 mL BAC Water → Concentration: 15.0 mg/mL

Dose (mg)	0.5mg	1.0mg	2.0mg	4.0mg	6.0mg	8.0mg
Volume (mL)	0.033	0.067	0.133	0.267	0.400	0.533
Syringe Units	3.3U	6.7U	13.3U	26.7U	40.0U	53.3U

Vial: 40mg x 10 — Add 2 mL BAC Water → Concentration: 20.0 mg/mL

Dose (mg)	0.5mg	1.0mg	2.0mg	4.0mg	6.0mg	8.0mg
Volume (mL)	0.025	0.050	0.100	0.200	0.300	0.400
Syringe Units	2.5U	5.0U	10.0U	20.0U	30.0U	40.0U

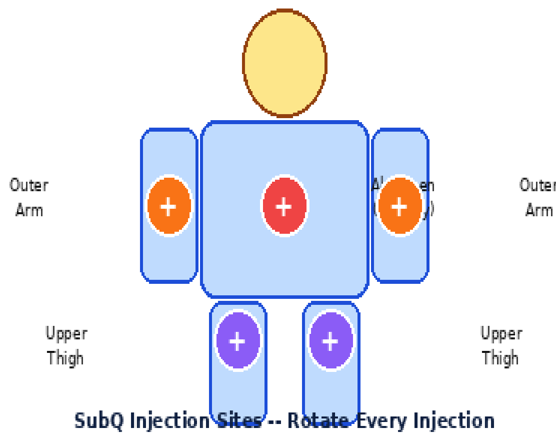
Vial: 60mg x 10 — Add 2 mL BAC Water → Concentration: 30.0 mg/mL

Dose (mg)	0.5mg	1.0mg	2.0mg	4.0mg	6.0mg	8.0mg
Volume (mL)	0.017	0.033	0.067	0.133	0.200	0.267
Syringe Units	1.7U	3.3U	6.7U	13.3U	20.0U	26.7U

Note: Doses above 4 mg are advanced research doses. Escalate only under qualified supervision. Always double-check your volume before drawing.

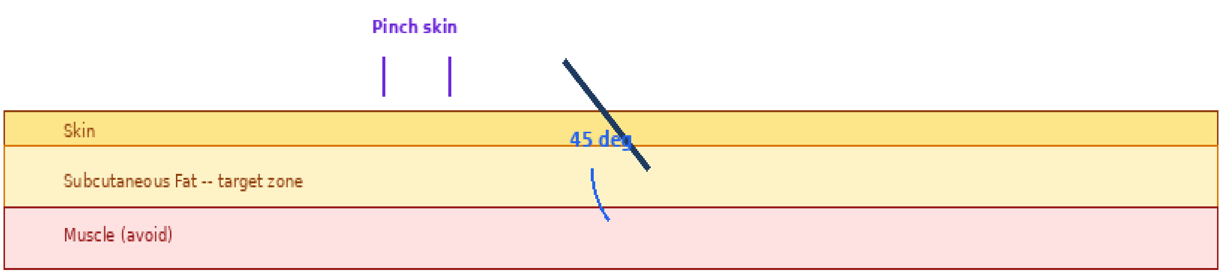
INJECTION TECHNIQUE

Step	Action	Detail
1	Wash hands	Wash with soap and warm water for 20 seconds. Dry completely.
2	Gather materials	Insulin syringe (29-31G x 5/16"), Retatrutide vial, alcohol swabs, sharps container.
3	Clean injection site	Wipe chosen site in circular motion with alcohol swab. Allow 10-15 seconds to dry completely.
4	Draw the dose	Draw correct volume per dose table. Tap and expel air bubbles.
5	Pinch and insert	Pinch 2-3 cm of skin. Insert at 45 degrees. Release pinch. Inject slowly over 5-10 seconds.
6	Withdraw and press	Remove at same 45-degree angle. Press site gently 5-10 sec. Do NOT rub — rubbing displaces the dose.
7	Safe disposal	Place needle immediately into sharps container. Never recap, reuse, or leave needles exposed.



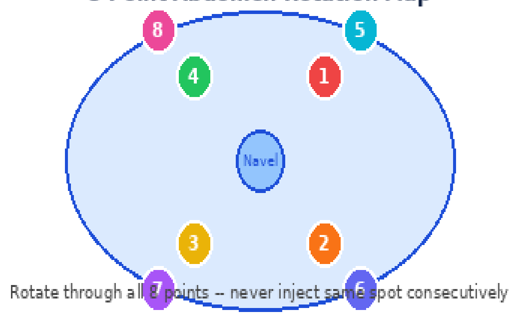
SubQ Injection Sites -- Rotate Every Injection

45-Degree SubQ Injection Angle



SITE ROTATION & DOSING PROTOCOL

8-Point Abdomen Rotation Map



Weekly Dosing Schedule -- Same Day Each Week

Choose any consistent day (Mon shown). 1 injection per week only.

Phase	Weeks	Dose	Frequency	Notes
Starter	1-4	0.5 mg	1x / week	Assess GI tolerance. Inject with or just after a meal.
Ramp-Up 1	5-8	1.0 mg	1x / week	Increase only if 0.5 mg is well tolerated.
Ramp-Up 2	9-12	2.0 mg	1x / week	Monitor weight, appetite, and energy.
Ramp-Up 3	13-16	4.0 mg	1x / week	Most research subjects reach maintenance here.
Maintenance	16+	4-8+ mg	1x / week	Titrate to response. Higher doses in Phase 2 trials.
Rest Period	Post-cycle	0 mg	4 weeks off	Reset receptor sensitivity before next cycle.

Administration tip: Inject with or shortly after a meal — this markedly reduces nausea, especially during dose escalation.

Consistency: Choose the same day each week. A 1-2 day variance is acceptable; never double-dose to compensate.

SIDE EFFECTS, STORAGE & GOLDEN RULES

Side Effect	Likelihood	Management
Nausea	Very Common	Inject with food. Small meals. Avoid fatty/spicy food on injection day.
Reduced Appetite	Very Common	Expected effect. Maintain adequate protein and micronutrient intake.
Vomiting	Common	Usually dose-related. Hold escalation until resolved.
Diarrhoea	Common	Typically transient. Stay hydrated. Reduce dose if persistent.
Constipation	Occasional	Increase fluid and fibre. Usually self-resolving.
Injection site reaction	Occasional	Rotate sites. Use fresh 29-31G needle each time.
Tachycardia / Palpitations	Rare	Stop dosing. Seek medical advice before resuming.

STOP & SEEK MEDICAL ADVICE: Severe persistent abdominal pain (rule out pancreatitis)

STOP & SEEK MEDICAL ADVICE: Neck lump, hoarseness, or difficulty swallowing (thyroid concern)

STOP & SEEK MEDICAL ADVICE: Severe allergic reaction: hives, facial swelling, breathing difficulty

STOP & SEEK MEDICAL ADVICE: Resting heart rate increase > 20 bpm above personal baseline

STOP & SEEK MEDICAL ADVICE: Jaundice or dark urine (hepatic concern)

Absolute Contraindications: Personal or family history of medullary thyroid carcinoma or MEN2 syndrome, active or history of pancreatitis, pregnancy or lactation. Not for use in individuals under 18.

Storage Guidelines

State	Temperature	Duration	Notes
Unreconstituted powder	2-8 deg C	Per expiry	Original vial. Dry, away from light.
Reconstituted solution	2-8 deg C	28 days	Label with date. Inspect before each use.
BAC Water (opened)	2-8 deg C	28 days	Label with opening date.
Short-term transport	Below 25 deg C	Max 72 hr	Insulated bag + ice packs. No direct ice contact.

Golden Rules — Retatrutide

1. Always start at 0.5 mg and escalate slowly — patience prevents side effects.
2. Inject with or after food to minimise nausea, especially during ramp-up.
3. One injection per week only — more frequent dosing does not improve outcomes.
4. Never inject the same site twice in a row — rotate through all 8 marked points.
5. Use only BAC Water for reconstitution. Never use saline or plain sterile water.
6. Swirl gently to mix — NEVER shake. Agitation denatures the peptide.
7. Label every vial with the reconstitution date. Discard after 28 days.
8. Store at 2-8 deg C at all times. Never freeze reconstituted peptide.
9. Contraindicated in thyroid cancer history, MEN2, pancreatitis, or pregnancy.

10. If GI side effects are severe, reduce dose before considering a full pause.